

is

$$w^* = \arg \min_w \sqrt{\sum_j \left(\frac{z_{w,j} - m_j}{s_j} \right)^2}. \quad (6.15)$$

This measured-feature medoid selects the actual window closest to the typical multivariate operating profile within each cell and SOH state. Model predictions and estimation errors do not enter the selection. The procedure therefore retains diversity across cells and ageing states without favoring one estimator.

The shared SOH-LSTM receives 192 h of undisturbed causal context before every scored interval. Standard scenarios use a 24-h evaluation window. The one-hour dropout uses 48 h from the same full-charge anchor so that sufficient post-gap behavior remains observable. Figure 6.3 summarizes the resulting selection and timing protocol.

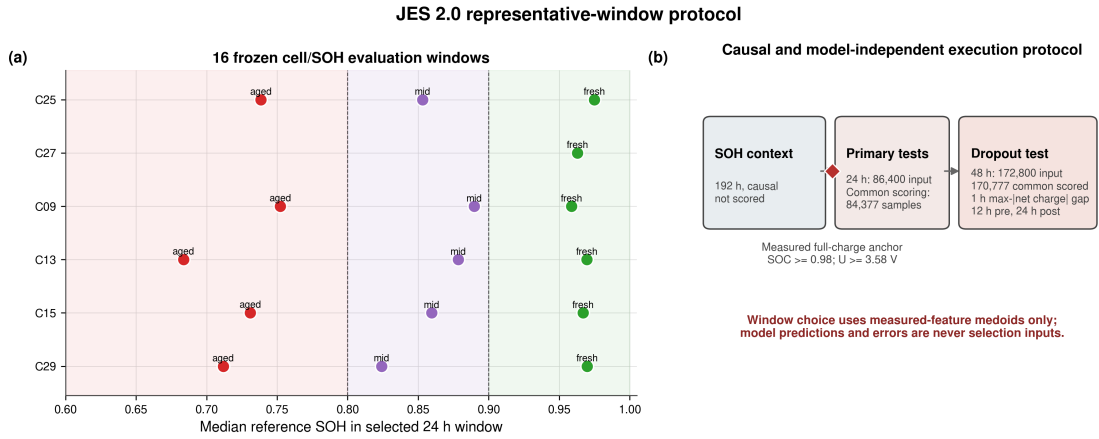


Figure 6.3: Representative-window protocol for the robustness benchmark. Sixteen cell/SOH windows are selected using measured-feature medoids. Every estimator is scored over the same interval after the causal SOH and rolling DD input contexts are available. Standard scenarios cover 24 h, while the dropout case uses 48 h and places its one-hour gap in the eligible interval with maximum absolute net charge.

The benchmark contains 20 scenario definitions. These comprise the nominal baseline, two current-noise levels, voltage and temperature noise, three signed current-gain magnitudes, one signed current-offset level, voltage and temperature offsets, ADC quantization, initial SOC mismatch, periodic and random sample loss, three timing-jitter levels, a one-hour burst dropout, and randomized-sign voltage spikes. Every current-gain magnitude and the 50 mA current-offset case contain matched positive and negative signs. Gaussian noise cases use 10 deterministic